Causal Forest in Top Journals: A Literature Review in Economics, Public Health, Genomics & Medicine

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Outline

Methodological Foundations

Economics Top Journals

Public Health Top Journals

Genomics & Precision Medicine

Medicine & Clinical Research

Methodology & Software Papers

Complete Reference List

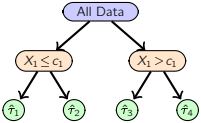
Methodological Foundations

Causal Tree — Athey & Imbens (2016)

PNAS

Athey, S. & Imbens, G. (2016). Recursive partitioning for heterogeneous causal effects. *PNAS*, 113(27), 7353–7360.

- Proposes **honest splitting**: separate samples for tree structure vs. effect estimation
- Modifies CART to optimize for treatment effect heterogeneity
- Provides valid CIs without sparsity assumptions



DOI: 10.1073/pnas.1510489113

Causal Forest — Wager & Athey (2018)

JASA

Wager, S. & Athey, S. (2018). Estimation and inference of heterogeneous treatment effects using random forests. *JASA*, 113(523), 1228–1242.

- Extends causal tree to **causal forest** via ensemble of honest trees
- Proves pointwise consistency and asymptotic normality
- Provides valid confidence intervals for CATE
- First random forest method with provably valid inference

DOI: 10.1080/01621459.2017.1319839

Generalized Random Forest — Athey, Tibshirani & Wager (2019)

Annals of Statistics

Athey, S., Tibshirani, J. & Wager, S. (2019). Generalized random forests. *Annals of Statistics*, 47(2), 1148–1178.

- Unifying framework: GRF solves **local moment equations** with adaptive forest-based weighting
- Subsumes causal forest, quantile forest, instrumental forest as special cases
- R package: `grf` — the standard implementation

DOI: 10.1214/18-AOS1709

RATE — Yadlowsky et al. (2021)

arXiv

Yadlowsky, S. et al. (2021). Evaluating treatment prioritization rules via rank-weighted average treatment effects. *arXiv:2111.07966*.

- RATE: metric to evaluate whether CATE estimates effectively prioritize treatment
- Two variants: **AUTOC** and **Qini**
- Implemented in `grf::rank_average_treatment_effect()`

arXiv: 2111.07966

Economics Top Journals

AER: Davis & Heller (2017)

American Economic Review

Davis, J.M.V. & Heller, S.B. (2017). Using causal forests to predict treatment heterogeneity: An application to summer jobs. *AER*, 107(5), 546–550.

- Two RCTs of a youth summer jobs program
- Identifies heterogeneity that standard interaction approaches miss

DOI: 10.1257/aer.p20171000

Econometrica: Athey & Wager (2021)

Econometrica

Athey, S. & Wager, S. (2021). Policy learning with observational data. *Econometrica*, 89(1), 133–161.

- Doubly robust policy learning with optimal rates
- Bridges causal forest estimation and optimal policy assignment
- Foundation for `policytree` R package

DOI: 10.3982/ECTA15732

Operations Research: Zhou, Athey & Wager (2023)

Operations Research

Zhou, Z., Athey, S. & Wager, S. (2023). Offline multi-action policy learning. *OR*, 71(1), 148–183.

- Extends policy learning to **multi-action** settings
- Sample-optimal algorithm for tree-based treatment policies
- Foundation for multi-arm causal forest

DOI: 10.1287/opre.2022.2350

Economic Journal: Knittel & Tanaka (2021)

Economic Journal

Knittel, C.R. & Tanaka, S. (2021). Using machine learning to target treatment: The case of household energy use. *Economic Journal*, 135(672), 2377–2409.

- Causal forest to target energy conservation nudges
- Forest increases social net benefits by factor of 3–5

DOI: 10.1093/ej/ueab019

Management Science

Management Science

Wang, G. (2022). Medicaid expansion and ED wait time. *Mgt. Sci.*, 68(9), 6648–6665.
DOI: 10.1287/mnsc.2022.4348

Serra-Garcia, M. & Szech, N. (2023). Incentives and defaults for COVID-19 vaccines. *Mgt. Sci.*, 69(2), 1037–1049.
DOI: 10.1287/mnsc.2022.4397

Labour Economics & More

Labour Economics

Cockx, B., Lechner, M. & Bollens, J. (2023). Priority to unemployed immigrants? *Labour Econ.*, 80, 102329.
DOI: 10.1016/j.labeco.2023.102329

J. Applied Econometrics

Gavrilova, E. & Langørgen, A. (2024). DiD causal forests with an application to payroll tax incidence in Norway. *J. Appl. Econ.*, 40(7), 727–740.
DOI: 10.1002/jae.70001

Environmental & Development Economics

Environmental Economics

Liu, C. (2023). Causal forest estimation of heterogeneous and time-varying environmental policy effects.
SSRN: 3447976

J. Econ. Behav. Org.

Dylong, P. & Uebelmesser, S. (2024). Biased beliefs about immigration. *JEBO*, 217, 453–482.
DOI: 10.1016/j.jebo.2023.11.020

J. Dev. Econ.

Mesplé-Somps, S. & Nilsson, B. (2023). Role models, aspirations and desire to migrate. *JDE*, 212, 819–839.
DOI: 10.1016/j.jdeveco.2023.103158

Health Economics

Health Economics

Heterogeneous effects of air pollution on pandemic patients. (2025). *Health Econ.*, 34(4), 655–676.
DOI: 10.1002/hec.70003

Econ. & Human Biology

Bodory, H. & Lechner, M. (2021). Regional differences in diabetes across Europe. *EHB*, 40, 100981.
DOI: 10.1016/j.ehb.2021.100981

Eur. J. Health Econ.

Hollenbach, J., Schmitz, H. & Westphal, M. (2026). Gene-environment interactions with essential heterogeneity. *EJHE*, forthcoming.
DOI: 10.1007/s10198-025-01675-2

Public Health Top Journals

American Journal of Epidemiology

AJE

Jawadekar, N. et al. (2023). Practical guide to honest causal forests. *AJE*, 192(7), 1155–1165.
DOI: 10.1093/aje/kwad043

Jakobsen, K.D. (2024). Harnessing causal forests for epidemiologic research. *AJE*, 193(6), 813–815.
DOI: 10.1093/aje/kwad219

Lee, Y. et al. (2024). Iterative causal forest for subgroup identification. *AJE*, 193(5), 811–825.
DOI: 10.1093/aje/kwad219

BMC Journals

BMC Medical Informatics

Dennis, J.M. et al. (2023). Causal forest vs. regression for type 2 diabetes. *BMC Med. Inform.*, 23, 143.
DOI: 10.1186/s12911-023-02207-2

BMC Med. Res. Methodol.

Application of causal forests to RCT data: a case study. (2025). *BMC MRM*, 25, 42.
DOI: 10.1186/s12874-025-02489-2

PLoS ONE & More Public Health

PLoS ONE

Hollenbach, J. et al. (2024). Heterogeneity within the Oregon Health Insurance Experiment.
DOI: 10.1371/journal.pone.0309883

Alcohol & Alcoholism

CF model for SUD psychosocial treatments. (2025). *Alcohol Alcohol.*
DOI: 10.1093/alcalc/agae077

J. Nutrition

Causal forest in FMNCP healthy food subsidy RCT. (2026). *J. Nutr.*
DOI: 10.1016/j.tjnut.2025.03.015

Genomics & Precision Medicine

Pharmacogenomics & Drug Response

Stat. Methods Med. Res.

Dandl, S. et al. (2024). What makes forest-based HTE estimators work? *SMMR*, 33(5).
DOI: 10.1177/09622802231224628

JRSS Series C

Case-only trees and random forests for gene–treatment interaction. *JRSS-C*, 68(5), 1371–1387.
DOI: 10.1111/rssc.12377

Pharmacoepidemiology

Lee, Y. et al. (2024). iCF for SGLT2i vs. GLP1RA subgroup identification. *AJE*, 193(5), 811–825.

Genomics & Polygenic Risk

Genomics Applications

Heterogeneous treatment effects of major depression polygenic risk scores on self-harm using causal forest. UK Biobank, n=91,778.
DOI: 10.1038/s41398-024-03136-3

Multi-Study CF

Venkatasubramanian, A. & Wolfson, J. (2025). Multi-study causal forest: A flexible framework for data borrowing. *arXiv:2502.02110*.
arXiv: 2502.02110

High-Dim PS + iCF

High-dimensional approach with iCF for SGLT2i HTE. (2025). *Pharmacoepidemiol. Drug Saf.*
DOI: 10.1002/pds.5924

Medicine & Clinical Research

Oncology & Surgery

Breast Cancer

Wang, Z. et al. (2023). Causal forest for TNBC postoperative radiotherapy. *Chin. J. Health Stat.*, 40(1), 32–35.

Stat. Methods Med. Res.

Seibold, H., Zeileis, A. & Hothorn, T. (2018). Model-based forests for HTE. *SMMR*, 27(10), 3104–3125.
DOI: 10.1177/0962280217693034

Biostatistics

Dorie, V. et al. (2019). Automated versus do-it-yourself methods for causal inference. *Biostatistics*, 20(4), 645–656.
DOI: 10.1093/biostatistics/kxy009

Critical Care & Sepsis

BMC Med. Res. Methodol.

Causal forests in VANISH RCT: vasopressin vs. norepinephrine for septic shock. (2025). *BMC MRM*, 25, 42.
DOI: 10.1186/s12874-025-02489-2

Ann. Int. Med.

Kent, D.M. et al. (2020). The predictive approaches to treatment effect heterogeneity (PATH) statement. *Ann. Intern. Med.*, 172(1), 35–45.
DOI: 10.7326/M18-3667

Mental Health & Psychiatry

Int. J. Methods Psychiatr. Res.

Sverdrup, E. et al. (2025). Estimating treatment effect heterogeneity in psychiatry: A tutorial with causal forests. *IJMPR*, 34(2).
DOI: 10.1002/mpr.70015

Substance Use Disorders

CF for SUD psychosocial treatments across 12 RCTs. (2025). *Alcohol Alcohol.*
DOI: 10.1093/alcalc/agae077

Precision Mental Health

Predicting HTE for depression through data integration. (2025). *arXiv:2509.04604.*
arXiv: 2509.04604

Methodology & Software Papers

Key Methodology Papers

Stat. Science & JASA

Nie, X. & Wager, S. (2021). Quasi-oracle estimation of HTE. *Biometrika*, 108(2), 299–319.
DOI: 10.1093/biomet/asaa076

Athey, S. & Imbens, G. (2019). Machine learning methods that economists should know about. *Ann. Rev. Econ.*, 11, 685–725.
DOI: 10.1146/annurev-economics-080217-053433

Lechner, M. (2019). Modified causal forests for estimating heterogeneous causal effects. *CEPR DP 13430*.
SSRN: 3300057

Software & Review Papers

Software

Tibshirani, J. et al. (2024). `grf`: Generalized Random Forests. R package.
grf-labs.github.io/grf

Sverdrup, E. et al. (2020). `policytree`: Policy learning via doubly robust EWM. *JOSS*, 5(50), 2232.
DOI: 10.21105/joss.02232

Bodory, H. et al. (2022). `mcf`: Modified Causal Forests. Python package.
mcfpy.github.io/mcf

Review

Rehill, P. (2024). How do applied researchers use the causal forest? *arXiv:2404.13356*.
arXiv: 2404.13356

Annals of Applied Statistics & More

AOAS

Dandl, S. et al. (2024). What makes forest-based HTE estimators work? *AOAS*, 18(2), 1350–1373.
DOI: 10.1214/23-AOAS1799

Int. Stat. Rev.

Rehill, P. (2025). How do applied researchers use the causal forest? *ISR*.
DOI: 10.1111/insr.12610

Entropy

Bodory, H., Busshoff, H. & Lechner, M. (2022). High resolution treatment effects estimation with MCF. *Entropy*, 24(8), 1039.
DOI: 10.3390/e24081039

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4. Cui, Kosorok, Sverdrup & Wager (2023). Causal survival forests. *JRSS-B*.
5. Nie & Wager (2021). Quasi-oracle estimation of HTE. *Biometrika*.
6. Yadlowsky et al. (2021). RATE. *arXiv:2111.07966*.
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12. Dorie et al. (2019). Automated vs do-it-yourself methods. *Biostatistics*.
13. Kent et al. (2020). PATH statement. *Ann. Intern. Med.*
14. Athey & Imbens (2019). ML methods economists should know. *Ann. Rev. Econ.*
15. Sverdrup et al. (2020). `policytree`. *JOSS*.
16. Bodory et al. (2022). High resolution with MCF. *Entropy*.
17. Rehill (2025). How do researchers use CF? *ISR*.

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17. Davis & Heller (2017). CF for summer jobs. *AER*.
18. Knittel & Tanaka (2021). ML to target household energy use. *Econ. J.*
19. Wang (2022). Medicaid expansion and ED wait time. *Mgt. Sci.*
20. Serra-Garcia & Szech (2023). COVID-19 vaccine incentives. *Mgt. Sci.*
21. Cockx, Lechner & Bollens (2023). Unemployed immigrants training. *Labour Econ.*
22. Gavrilova & Langørgen (2024). DiD causal forests, payroll tax. *J. Appl. Econ.*
23. Liu (2023). Heterogeneous environmental policy effects. *SSRN*.
24. Dylong & Uebelmesser (2024). Immigration beliefs. *JEBO*.
25. Mesplé-Somps & Nilsson (2023). Role models and migration. *JDE*.
26. Air pollution HTE (2025). *Health Econ.*
27. Bodory & Lechner (2021). Diabetes across Europe. *EHB*.
28. Hollenbach et al. (2026). Gene-environment interactions. *EJHE*.

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35. CF for SUD treatments (2025). *Alcohol Alcohol.*
36. CF in FMNCP food subsidy RCT (2026). *J. Nutr.*
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40. Venkatasubramanian & Wolfson (2025). Multi-study CF. *arXiv*.
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Thank You!

Questions & Discussion

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